

MeshNet

Offline-First Mesh Communication for
Network Blackout Scenarios

Ishani, Tanvi, Yashika, Himasree



System Design



01

Messages stored locally on each device using IndexedDB/localStorage. Zero server dependency.

02

QR-based burst transfer, payload compressed into animated frames, scanned device-to-device.

03

Multi-hop propagation — each node relays messages further, hop count tracked per message.

04

PWA with Service Worker — full offline load after first visit, installs to home screen.

Key Features



1

Local Feed

compose and read mesh messages by type (Alert, Safe Route, Medical, DM)

2

Nearby Tab

live animated QR burst transfer between two people in proximity

3

QR Drop

static QR for billboard/wall dead drops, scannable anytime

4

Direct Messages

AES-GCM encrypted, only recipient can decrypt, safe on seized devices

5

Emergency Alerts

priority flag, up to 10 hops, no expiry, visible across all nodes

Design Decisions

Privacy Over Identity

No usernames. Node IDs are random. Sender name stripped from payload.

Reach Over Bandwidth

Multi-frame QR bursts sacrifice speed for larger payload capacity.

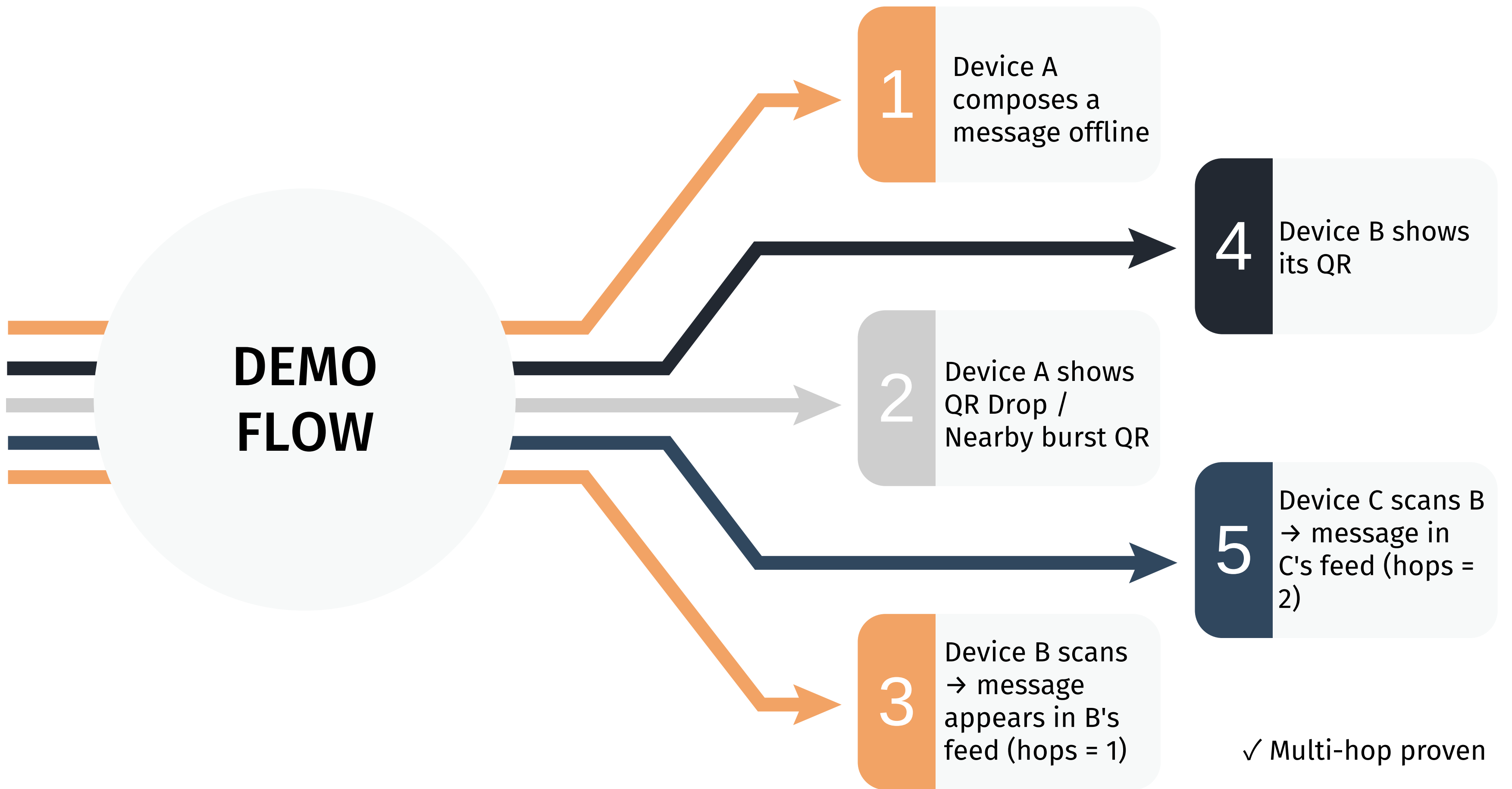


Resilience Over Consistency

No central coordinator. Each node is independent and replaceable.

Simplicity over Native

Web PWA runs on any phone browser, no app install required.



The Team



Ishani Singh

the voice of the
team, the heart of
the group



Tanvi Borkar

the calm in the
chaos, the hype
when it matters



Yashika Gupta

thinks twice,
delivers every time



Himasree D

chronically online,
surprisingly
functional



Thankyou